

Voicelsolate Pro

v25.0.0 — Current State Product Snapshot

Generated 2026-07-28 · 100% on-device voice isolation

This document describes the product **as of v25.0.0**: architecture contracts, Engineer Mode slider discipline, platforms, versions, verification status, and contributor rules. Linked from the repository README and docs index.

Field	Value
Product version	25.0.0
Android versionName / versionCode	25.0.0 / 250000
iOS CFBundleShortVersionString / CFBundleVersion	25.0.0 / 250000
Electron artifact pattern	Voicelsolate-Pro-25.0.0-win-x64.exe
Package manager	pnpm ≥ 11 · Node ≥ 22
Privacy	100% local · no cloud audio
Spectral contract	Exactly one Forward STFT + one Inverse STFT

1. Overview

Voicelso Pro extracts studio-quality speech from noisy audio and video **entirely on-device**. Surfaces: Landing Stem-Split (/), Engineer Mode (/app/), and Downloads. Upload-only workflow — live microphone capture is intentionally disabled.

1.1 Workflow

- **Upload** — accept File only (no PCM decode freeze).
- **Analyze** — decode, map speech/noise/music/whisper/impulse; joint Analyzer↔WhisperHunter map.
- **Process** — offline ML isolation (once per file) + single-pass spectral cleanup.
- **Live-Mix** — real-time gate/de-esser/EQ/comp via AudioWorklet; sliders never re-run ML.
- **Export** — WAV (and video re-mux where applicable), forensic audit log.

1.2 Hard constraints (non-negotiable)

Constraint	Rule
Privacy	No user audio or derivatives leave the device.
Single STFT / iSTFT	One forward + one inverse per offline spectral path.
No live mic	getUserMedia forbidden; Permissions-Policy microphone=().
Sliders ≠ re-inference	UI controls GainNodes / AudioParams only after stems exist.
Models	Same-origin /app/models/*.onnx or desktop seed — no CDN scripts.
Worklet ring buffer	Do not change voice-isolate-processor.js pointer math without explicit task.

2. Engineer Mode v25 — Slider discipline

v25 is a **hardening / polish** pass — not an architecture rewrite. Visible slider ranges stay as displayed; safer behavior lives in the **mapping / calibration** layer only.

2.1 Calibration curves

- **voicelso**: raw 0–72 ≈ linear (default 72 preserved). Above 72: cubic ease-out; UI 100 → effective ≈ 86.
- **bgSuppress**: linear to 60; upper range log-taper (max effective ≈ 88).
- **crosstalkCancel**: power-curve soft-start; further scaled by stereo channel-difference gate.
- API: pure `calibrate(sliderId, rawValue)` and `getEffectiveDspParams(rawParams)` in `slider-calibration.js`.

2.2 Coupling and soft clamps

- High effective voicelso can cap bgSuppress unless a speech-safe band (≈800–3400 Hz coverage) exists.
- Protected speech window: minimum band width enforced on effective edges only.
- Soft clamp de-risks extreme iso + suppress + narrow band (gargling / pumping risk).
- Dev logging: `set globalThis.VIP_DEBUG_CALIBRATION = true`.

2.3 Lock UI

- Every slider row: SVG padlock, data-locked, cyan protected accent (not red, not disabled gray).
- `toggleSliderLock(id)`; persist vip-slider-locks in `localStorage`.
- Locked: no drag, no preset overwrite, no reset unless full reset chosen; Reset Unlocked Only available.

2.4 Metrics, sections, overlay

- Single `updateAudioMetrics()` for Voice %, Noise %, SNR dB (header + strip + neon pulse).
- Collapsible native details/summary for major UI sections.
- Processing overlay variants: uploading → decoding → analyzing → separating → isolating → reconstructing → exporting; always hidden in `runPipeline` finally.

3. Platforms and version matrix

Surface	Technology	Version sync
Web	Vercel + Express local	package.json 25.0.0
Android	Capacitor + Gradle	versionName 25.0.0 · versionCode 250000
Desktop	Electron + electron-builder	artifact Voicelsolate-Pro-25.0.0-*
iOS (secondary v1 scope)	Capacitor plist	25.0.0 / 250000 when built

Run `pnpm mobile:sync-version` after any `package.json` version bump. GitHub Release binaries are published separately; until a v25 release is uploaded, latest may still resolve to prior v24 assets.

4. Architecture (summary)

Stem-Split and Live-Mix: Phase 1 offline ML inference once per file → stems; Phase 2 real-time Live-Mix graph (gains/filters only). New code in `src/` four-layer ESM; Engineer shell remains under `public/app/` in maintenance with explicit v25 polish in scoped files. Authoritative contributor contract: `CLAUDE.md`.

5. Key modules (v25 touch list)

Path	Role
<code>public/app/slider-calibration.js</code>	Discipline curves, coupling, soft clamps
<code>public/app/slider-map.js</code>	Registry, STAGES, structured SLIDER_HINTS
<code>public/app/slider-hint-ui.js</code>	Hint panels + expandable metadata
<code>public/app/app.js</code>	Locks, metrics, effective params, pipeline wiring
<code>public/app/index.html</code>	Collapsibles, metric DOM, reset unlocked
<code>public/app/style.css</code>	Lock/collapsible/hint/overlay styling
<code>public/app/processing-overlay.js</code>	Stage variants + hide-on-finally patch
<code>tests/slider-calibration-hardening.test.js</code>	Curve / clamp / lock contract tests

6. Verification status

- `pnpm test`: full Jest suite green after v25 hardening (2400+ tests).
- New unit coverage: calibration points (incl. 72 and 100), coupling clamp/non-clamp, soft clamp, lock persistence.
- `pnpm test:live`: Playwright smoke may still fail a pre-existing peak > 1.001 assertion; not a v25 mapping regression.
- ESLint: class methods use `document.querySelectorAll` (not `bindEvents-local qsa`).

7. Spelling and UI copy checklist

- Product name: **Voicelsolate Pro** (one word Voicelsolate, space Pro).

- Engineer Mode labels: Voice Isolation, BG Suppress, Voice Focus Lo/Hi, Crosstalk Cancel.
- Controls: Reset Controls · Reset Unlocked Only · Process · Reprocess · Forensic.
- Metrics labels: Voice % · Noise % · SNR Gain / SNR.
- Section summaries: Upload · Presets and Processing Controls · Processing Pipeline · Waveform and Spectrum · Noise Reduction / Gate · EQ · Dynamics · Spectral · Advanced · Output · Separation · EXTREME.
- Contributor doc spelling: CLAUDE.md (not CloudMD).

8. Commands

pnpm install · pnpm dev · pnpm test · pnpm test:live · pnpm mobile:sync-version · pnpm android:build:win · pnpm build:electron · pnpm validate · pnpm worklets:verify

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